

ACB – AIR CIRCUIT BREAKER

AUTO CHANGEOVER SCHEME, CONTROL LOGIC & INTERLOCKING

1. INTRODUCTION

ACB stands for Air Circuit Breaker.

An Air Circuit Breaker is a low-voltage circuit breaker used for switching, protection, isolation and control of electrical power systems.

ACBs are commonly used in:

- Main Incoming Feeders
- Bus Coupler / Bus Tie
- Outgoing Feeders
- Transformer Incomers
- Generator Incomers
- Main LT Panels
- PCC Panels
- MCC Panels
- Automatic Changeover Systems

An ACB can be operated manually as well as electrically, depending on the selected breaker type and control arrangement.

2. TYPICAL TWO-INCOMER ACB ARRANGEMENT

A typical two-source ACB panel consists of:

Incomer-1 ACB

Connected between Source-1 and Bus-1.

Incomer-2 ACB

Connected between Source-2 and Bus-2.

Bus Coupler ACB

Connected between Bus-1 and Bus-2.

Basic arrangement:

SOURCE-1 → INCOMER-1 ACB → BUS-1

SOURCE-2 → INCOMER-2 ACB → BUS-2

BUS-1 → BUS COUPLER ACB → BUS-2

Outgoing feeders are connected to the respective bus sections as per the project requirement.

3. WHAT IS AUTO CHANGEOVER?

An Auto Changeover Scheme is used to transfer the electrical load from one bus/source to another bus/source under predefined conditions.

The uploaded reference scheme states that the load can be transferred from one bus to another bus by automatically switching the breakers ON or OFF under specified conditions.

The purpose of an Auto Changeover Scheme is to:

- Maintain continuity of power supply
- Transfer load to an available healthy source
- Reduce interruption to the load
- Automatically operate the Bus Coupler
- Prevent incorrect breaker operation
- Provide controlled switching between two sources

4. MODES OF CHANGEOVER

The Auto Changeover Scheme consists of the following operating modes:

1. Independent Mode
2. Manual / Synchronizing Mode
3. Auto Mode
4. Blackout Condition

These modes are selected through the Mode Selector Switch.

5. MODE SELECTOR SWITCH

The Mode Selector Switch is provided on the Bus Coupler section.

The switch is a:

3-Pole, 3-Position Switch

The three positions are:

Position-1 – AUTO

Used for automatic changeover operation.

Position-2 – INDEPENDENT

Used for independent operation of the two bus sections.

Position-3 – MANUAL

Used for manual/synchronizing operation.

The same three positions are specified in the uploaded reference scheme.

6. TRIP SELECTOR SWITCH

The Trip Selector Switch is provided on the Bus Coupler section.

It is a:

3-Pole, 3-Position Switch

The three positions are:

1. Incomer-1 Trip
2. Incomer-2 Trip
3. Bus Coupler Trip

The Trip Selector Switch determines which breaker is selected for the trip operation during the manual/synchronizing sequence.

7. INDEPENDENT MODE

In Independent Mode, the two bus sections can operate independently.

In this mode, two breakers can be ON at the same time.

Typical operating conditions are:

Condition-1

Incomer-1 = ON

Incomer-2 = ON

Bus Coupler = OFF

Both incomers supply their respective bus sections.

Condition-2

Incomer-1 = ON

Incomer-2 = OFF

Bus Coupler = ON

Source-1 can supply both bus sections through the Bus Coupler.

Condition-3

Incomer-1 = OFF

Incomer-2 = ON

Bus Coupler = ON

Source-2 can supply both bus sections through the Bus Coupler.

These operating combinations are given as examples in the uploaded scheme.

8. INDEPENDENT MODE – INCOMER-1 CLOSING CONDITIONS

The closing conditions for Incomer-1 in Independent Mode are:

1. Mode Selector Switch must be in Independent Mode.
2. Bus PT-1 must be unhealthy.
3. Incomer-2 must be OFF.
4. Incomer-2 must not have a fault trip condition.
5. Bus Coupler must be OFF.
6. Bus Coupler must not have a fault trip condition.

When these conditions are satisfied, the closing operation of Incomer-1 can be permitted.

9. INDEPENDENT MODE – INCOMER-2 CLOSING CONDITIONS

The closing conditions for Incomer-2 are:

1. Mode Selector Switch must be in Independent Mode.

2. Bus PT-2 must be unhealthy.
3. Incomer-1 must be OFF.
4. Incomer-1 must not have a fault trip condition.
5. Bus Coupler must be OFF.
6. Bus Coupler must not have a fault trip condition.

These conditions are specified in the reference scheme.

10. INDEPENDENT MODE – BUS COUPLER CLOSING CONDITIONS

The Bus Coupler closing conditions are:

1. Mode Selector Switch must be in Independent Mode.
2. Bus PT-1 must be unhealthy.
3. Bus PT-2 must be unhealthy.
4. Incomer-1 must be OFF.
5. Incomer-1 must not have a fault trip condition.
6. Incomer-2 must be OFF.
7. Incomer-2 must not have a fault trip condition.

When these conditions are satisfied, Bus Coupler closing can be permitted.

11. MANUAL / SYNCHRONIZING MODE

In Manual Mode, when the two supplies are synchronized, the three breakers can be closed according to the defined operating sequence.

The three breakers are:

- Incomer-1
- Incomer-2
- Bus Coupler

After the required time delay, one selected breaker can be tripped according to the Trip Selector Switch setting.

The reference scheme specifies that when the supplies are synchronized, three breakers can close and one breaker is subsequently tripped after a time delay.

12. SYNCHRONIZING

Synchronizing is the process of checking the electrical conditions of two sources before they are connected together.

The reference scheme states that if synchronizing is required, the synchronizing function should be available in the numerical relay of the Bus Coupler.

Typical synchronizing parameters may include:

- Voltage
- Frequency

- Phase sequence
- Phase angle

Important: The uploaded document does not specify exact synchronizing settings or permissible values. Therefore, the actual settings must be taken from the approved project protection/control philosophy and relay manufacturer's documentation.

13. MANUAL MODE – INCOMER-1 CLOSING

The closing conditions for Incomer-1 are:

1. Mode Selector Switch = Manual Mode
2. Supply = Synchronized
3. Incomer-2 = ON and In Service
4. Bus Coupler = ON and In Service
5. Trip Selector Switch = Bus Coupler
6. Trip Selector Switch = Incomer-2

These conditions are given in the uploaded scheme.

14. MANUAL MODE – INCOMER-1 TRIPPING

The tripping conditions for Incomer-1 are:

1. Trip Selector Switch = Incomer-1

2. Incomer-2 = ON and In Service
3. Bus Coupler = ON and In Service
4. After the specified time delay, Incomer-1 will trip.

15. MANUAL MODE – INCOMER-2 CLOSING

The closing conditions for Incomer-2 are:

1. Mode Selector Switch = Manual Mode
2. Supply = Synchronized
3. Incomer-1 = ON and In Service
4. Bus Coupler = ON and In Service
5. Trip Selector Switch = Bus Coupler
6. Trip Selector Switch = Incomer-1

16. MANUAL MODE – INCOMER-2 TRIPPING

The tripping conditions for Incomer-2 are:

1. Trip Selector Switch = Incomer-2
2. Incomer-1 = ON and In Service
3. Bus Coupler = ON and In Service
4. After the specified time delay, Incomer-2 will trip.

17. MANUAL MODE – BUS COUPLER CLOSING

The closing conditions for the Bus Coupler are:

1. Mode Selector Switch = Manual Mode
2. Supply = Synchronized
3. Incomer-1 = ON and In Service
4. Incomer-2 = ON and In Service
5. Trip Selector Switch = Incomer-1
6. Trip Selector Switch = Incomer-2

18. MANUAL MODE – BUS COUPLER TRIPPING

The tripping conditions for the Bus Coupler are:

1. Trip Selector Switch = Bus Coupler
2. Incomer-1 = ON and In Service
3. Incomer-2 = ON and In Service
4. After the specified time delay, the selected breaker will trip according to the scheme.

The reference scheme also states that if both Incomer-1 and Incomer-2 become OFF under any condition, the Bus Coupler will also become OFF.

19. AUTO MODE

Auto Mode is provided for automatic Bus Coupler operation.

When one bus section becomes unhealthy due to an under-voltage condition, the control system checks the availability of the healthy source and the required breaker conditions.

If all conditions are satisfied, the Bus Coupler is automatically closed.

The reference document specifically describes Auto Mode for Bus Coupler operation.

20. AUTO MODE – BUS SIDE-A FAILURE

When an Under Voltage (U/V) condition occurs on Bus Side-A, the following conditions are checked:

1. Selector Switch = Auto Mode
2. Incomer-1 = OFF
3. Incomer-1 has no fault trip
4. Incomer-2 = ON and In Service
5. Bus PT-1 = Unhealthy
6. Bus PT-2 = Healthy

When all the above conditions are satisfied, the contactor picks up and the Bus Coupler breaker is closed.

This allows the healthy source connected through Incomer-2 to supply the affected bus section through the Bus Coupler.

21. AUTO MODE – BUS SIDE-B FAILURE

When an Under Voltage (U/V) condition occurs on Bus Side-B, the following conditions are checked:

1. Selector Switch = Auto Mode
2. Incomer-2 = OFF
3. Incomer-2 has no fault trip
4. Incomer-1 = ON and In Service
5. Bus PT-2 = Unhealthy
6. Bus PT-1 = Healthy

When all the above conditions are satisfied, the contactor picks up and the Bus Coupler breaker is closed.

This allows the healthy source connected through Incomer-1 to supply the affected bus section through the Bus Coupler.

22. BUS PT

Bus PT means Bus Potential Transformer.

The Bus PT is used for:

- Measuring bus voltage
- Monitoring bus voltage
- Checking bus healthy condition
- Checking bus unhealthy condition

The reference scheme specifically states that Bus PT is used for measuring bus voltage and checking whether the bus is healthy or unhealthy.

Typical arrangement:

Bus PT-1 → Bus-1 Voltage Monitoring

Bus PT-2 → Bus-2 Voltage Monitoring

23. LINE PT

Line PT means Line Potential Transformer.

Line PT is used for:

- Measuring line voltage
- Monitoring source availability
- Synchronizing two sources

The reference document specifies Line PT-1 and Line PT-2 and states that Line PT is used for line voltage measurement and synchronization between two sources.

24. NUMERICAL RELAY

The numerical relay is an important component of the control and protection system.

The reference scheme specifies that numerical relays should have maximum available:

- BI – Binary Inputs
- BO – Binary Outputs

If synchronizing is required, the Bus Coupler numerical relay should have a synchronizing function.

Binary Input (BI)

Binary inputs are used to receive status signals from:

- ACB
- Selector switches
- Protection relays
- PT monitoring
- Auxiliary contacts
- Other field devices

Binary Output (BO)

Binary outputs can be used to provide commands such as:

- ACB Close
- ACB Trip
- Bus Coupler Close
- Bus Coupler Trip
- Alarm/Control commands

Exact BI/BO allocation must be finalized from the approved control schematic.

25. ACB PROTECTION FUNCTIONS

Depending on the selected ACB trip unit and project requirement, typical ACB protection functions may include:

- Overload Protection
- Short Circuit Protection
- Instantaneous Protection
- Earth Fault Protection
- Under Voltage Protection
- Over Voltage Protection
- Under Frequency Protection
- Over Frequency Protection
- Other protection functions as required

The exact protection functions and settings depend on the selected ACB, trip unit and approved protection coordination study.

26. ACB AUXILIARY CONTACTS

Auxiliary contacts are used to provide breaker status feedback to the control system.

Typical breaker feedback signals include:

- ACB ON
- ACB OFF
- ACB TRIP
- Ready to Close
- Spring Charged
- Service Position
- Test Position
- Fault Trip
- Local/Remote Status

Typical auxiliary contacts may include:

52a

Indicates breaker closed status.

52b

Indicates breaker open status.

The actual contact numbering must be confirmed from the selected ACB manufacturer's wiring diagram.

27. ACB ELECTRICAL OPERATION

An electrically operated ACB may have the following control functions:

CLOSE COMMAND

Used to electrically close the ACB.

TRIP / OPEN COMMAND

Used to open or trip the ACB.

SPRING CHARGED

Indicates that the closing mechanism is charged and ready.

READY TO CLOSE

Indicates that the breaker is available for closing when all permissive conditions are satisfied.

TRIP STATUS

Indicates that the breaker has operated due to protection or an external trip command.

Exact control terminals and coil ratings depend on the selected ACB manufacturer and model.

28. ELECTRICAL INTERLOCK

Electrical interlocking is used to prevent incorrect breaker operation through the control circuit.

Typical purposes include:

- Preventing unwanted paralleling of two sources
- Preventing incorrect Bus Coupler operation
- Preventing closing of a faulted breaker
- Preventing breaker closing when the Earth Switch is ON
- Preventing incorrect isolator operation
- Ensuring correct changeover sequence

The final electrical interlock circuit must be based on the approved control schematic.

29. MECHANICAL INTERLOCK

Mechanical interlocking uses physical mechanisms to prevent incorrect operation.

It can be used between:

- ACBs
- Bus Coupler
- Isolators
- Earth Switches
- Draw-out breaker mechanisms

Mechanical interlocking provides an additional level of protection against incorrect manual operation.

30. IMPORTANT INTERLOCKS

The reference scheme provides several important interlocking requirements.

30.1 Door Interlock

It shall not be possible to open the front door when the circuit breaker is ON and in the service position.

30.2 Breaker Racking Interlock

It shall not be possible to rack the breaker OUT from Service to Test position when the breaker is ON.

30.3 Test-to-Service Interlock

It shall not be possible to rack the breaker IN from Test to Service position when the breaker is ON.

30.4 Chamber Door Interlock

It shall not be possible to rack IN the breaker when the front breaker chamber door is open.

A suitable defeat interlock mechanism may be provided for emergency purposes.

30.5 Earthing Switch Interlock

The earthing switch shall only be switched ON when the withdrawable truck is in the Test position, where applicable.

30.6 Earth Switch ON Interlock

The breaker shall not be racked IN from Test to Service position when the earthing switch is ON.

30.7 Isolator Interlock

The circuit breaker shall not be switched ON when the isolator is OPEN.

The isolator shall not be opened when the circuit breaker is ON.

30.8 Earth Switch Interlock

The earth switch shall not be closed when the circuit breaker is ON.

The earth switch shall not be closed when the isolator is CLOSED.

30.9 Earth Switch ON Condition

The circuit breaker and isolator shall not be closed when the earth switch is ON.

31. TYPICAL NORMAL OPERATING CONDITION

Under normal operation:

Incomer-1 ACB = ON

Incomer-2 ACB = ON

Bus Coupler ACB = OFF

In this condition, both bus sections receive power from their respective incomers.

32. SOURCE-1 FAILURE – AUTO CHANGEOVER SEQUENCE

Normal condition:

I/C-1 = ON

I/C-2 = ON

B/C = OFF

When Source-1 fails:

1. Bus PT-1 detects unhealthy bus condition.
2. Incomer-1 becomes/ is confirmed OFF.
3. Incomer-1 must not have a fault-trip condition.
4. Incomer-2 must be ON and in service.
5. Bus PT-2 must indicate healthy bus condition.
6. Auto Mode must be selected.
7. Required interlocks must be healthy.
8. Bus Coupler closing command is generated.
9. Bus Coupler closes.

The above sequence follows the Auto Mode conditions stated in the reference scheme.

33. SOURCE-2 FAILURE – AUTO CHANGEOVER SEQUENCE

Normal condition:

I/C-1 = ON

I/C-2 = ON

B/C = OFF

When Source-2 fails:

1. Bus PT-2 detects unhealthy bus condition.
2. Incomer-2 becomes/ is confirmed OFF.
3. Incomer-2 must not have a fault-trip condition.
4. Incomer-1 must be ON and in service.
5. Bus PT-1 must indicate healthy bus condition.
6. Auto Mode must be selected.
7. Required interlocks must be healthy.
8. Bus Coupler closing command is generated.
9. Bus Coupler closes.

The above sequence follows the Auto Mode conditions stated in the reference scheme.

34. AUTO CHANGEOVER PERMISSIVES

Before giving a Bus Coupler closing command, the following conditions should be verified as applicable to the approved scheme:

1. Auto Mode selected.
2. Healthy source available.
3. Failed source breaker OFF.
4. Healthy source breaker ON and in service.
5. Bus voltage status available.
6. No unwanted fault-trip condition.
7. Bus Coupler available.

8. Required interlocks healthy.
9. Breaker closing permissive available.
10. Synchronizing condition available if required.

35. ACB DRAW-OUT POSITIONS

For a draw-out type ACB, typical positions may include:

SERVICE / CONNECTED POSITION

The main power contacts are connected and the breaker is in service.

TEST POSITION

The main power circuit is isolated while the control circuit may remain available for testing, depending on the ACB design.

DISCONNECTED / ISOLATED POSITION

The breaker is isolated from the main power circuit.

Exact position names and functionality shall be confirmed from the selected ACB manufacturer's documentation.

36. LOCAL AND REMOTE OPERATION

LOCAL OPERATION

The ACB is operated from the local panel/control mechanism.

REMOTE OPERATION

The ACB may be operated through:

- PLC
- Numerical Relay
- SCADA
- BMS
- Remote Control Panel

depending on the project control philosophy.

For Auto Changeover, remote/automatic operation must only be permitted when all required interlocks and permissive conditions are healthy.

37. BLACKOUT CONDITION

Blackout condition refers to a situation where the required power sources are unavailable.

The uploaded scheme lists Blackout Condition as one of the changeover modes, but it does not provide a complete detailed blackout operating sequence.

Therefore, the exact blackout sequence must be defined by the project-specific control philosophy.

38. FOUR-INCOMER SYSTEM

If the panel contains:

4 Nos. Incomer Feeders

and

1 No. Bus Coupler

additional interlocking may be required.

The uploaded reference scheme specifically states that additional interlocks shall be provided when four incomer feeders and one Bus Coupler are available.

39. DESIGN CHECKLIST FOR ACB PANEL

Electrical Parameters

- Rated Voltage
- Rated Current
- Frequency

- Number of Poles
- Short Circuit Breaking Capacity
- Short Time Withstand Current
- Neutral Requirement
- Protection Functions
- Trip Unit
- Control Voltage

Mechanical Parameters

- Fixed or Draw-out Type
- ACB Compartment
- Racking Mechanism
- Service Position
- Test Position
- Isolated Position
- Door Interlock
- Mechanical Interlock
- Shutter Arrangement
- Busbar Connection
- Cable Connection

Control Parameters

- Close Command
- Trip Command
- ON Feedback
- OFF Feedback

- Trip Feedback
- Spring Charged
- Ready to Close
- Service Position
- Test Position
- Local/Remote
- PLC Interface
- SCADA Interface

40. ACB VS VCB

The uploaded reference document contains some interlocking points specifically referring to a Vacuum Circuit Breaker (VCB) and its trolley/racking arrangement.

ACB

Air Circuit Breaker

Generally used for low-voltage applications.

VCB

Vacuum Circuit Breaker

Generally used for medium-voltage applications.

Therefore, the following portions of the reference scheme can be used as the basis for an ACB Auto Changeover philosophy:

- Auto Changeover Logic
- Bus PT Logic
- Line PT Logic
- Selector Switch Logic
- Breaker Status Logic
- Electrical Interlocking Philosophy

However, VCB-specific mechanical/racking details should not be directly copied into an ACB drawing without checking the selected ACB construction and manufacturer's documentation.

41. TYPICAL ACB SIGNAL LIST

A typical ACB control system may use the following signals:

Signal | Description

ACB ON | Breaker closed status

ACB OFF | Breaker open status

ACB TRIP | Breaker trip status

Fault Trip | Protection trip status

Spring Charged | Closing mechanism charged

Ready to Close | Breaker closing permissive

Service Position | Breaker i